Hints: Continuous functions – 2

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Note. The value is not determined by the hypothesis: one can only prove $\operatorname{osc} f(0) \geqslant 1$, and every value in $[1, +\infty]$ occurs. The hints prove both facts.

Hint 1. Every neighbourhood U of 0 contains a neighbourhood V of some point $x \neq 0$, and $\sup_U f - \inf_U f \geqslant \sup_V f - \inf_V f \geqslant 1$. So $\operatorname{osc} f(0) \geqslant 1$.

Hint 2. Examples: let D be the Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere), which has oscillation 1 everywhere. Add a function g that is continuous on $\mathbb{R} \setminus \{0\}$, e.g. $g(x) = c \sin \frac{1}{x}$ (oscillation $1 + 2c$ at 0) or $g(x) = \frac{1}{x}$ (oscillation $+\infty$), with any value at 0.

Problem 2

Hint 1. Answer: no. The condition implies, for each n, that the closed sets $E_n = \{x : |f(y) - f(x)| \geqslant |y - x| \text{ whenever } |y - x| \leqslant \frac{1}{n}\}$ cover $\mathbb{R}$.

Hint 2. By Baire's theorem some E_n contains an interval J of length $< \frac{1}{n}$. On J, f is injective, hence strictly monotone; say increasing. Then $(f(y) - f(x))/(y - x) \to +\infty$ at every point of J.

Hint 3. Take $[c, d] \subset J$ and $g(x) = f(x) - Kx$ with $K > \frac{f(d) - f(c)}{d - c}$, so that $g(d) < g(c)$. But at every point $g(y) > g(x)$ for y slightly to the right of x, so the maximum of g on $[c, d]$ can only be attained at d. Contradiction.

Problem 3

(a)

Hint 1. False: $f(x) = x^3 - x$.

(b)

Hint 1. True: a monotone function whose range is an interval has no jumps (Theorem 2).

(c)

Hint 1. False: $f = \arctan$, or a constant.

Problem 4

Hint 1. One direction is trivial. Conversely, suppose $x_{n+1} - x_n \to 0$ but $a = \liminf x_n < \limsup x_n = b$. Since the steps tend to 0, every point of $[a, b]$ is a limit point of the sequence.

Hint 2. Show that $f(c) = c$ for every $c \in (a, b)$: otherwise $|f(x) - x| > \eta > 0$ near c, while the sequence visits that neighbourhood infinitely often with arbitrarily small steps $|f(x_n) - x_n|$.

Hint 3. So f is the identity on (a, b), and once x_n enters (a, b) (which it does) the sequence becomes constant. Contradiction.

Problem 5

Hint 1. Move the fixed point $-\frac{1}{6}$ of $x \mapsto 7x + 1$ to the origin: $g(t) = f(t - \frac{1}{6})$ satisfies $g(7t) = 49g(t)$.

Hint 2. Differentiate twice: $g''(7t) = g''(t)$, so $g''(t) = g''(t/7^n) \to g''(0)$. Hence g is a quadratic polynomial.

Hint 3. Substituting $g = At^2 + Bt + C$ forces $B = C = 0$. Answer: $f(x) = c\,(6x+1)^2$.

Problem 6

(a)

Hint 1. $f(x) = x \sin \frac{1}{x}$, $f(0) = 0$, crosses the axis at every point $\frac{1}{k\pi}$.

(b)

Hint 1. Answer: yes. Use the Cantor set C: let $f = 0$ on C and put bumps of alternating signs on the removed intervals.

Hint 2. Say that a removed interval has level k if it was removed at step k. Define $f(x) = (-1)^k \operatorname{dist}(x, C)$ on removed intervals of level k. Then $|f| = \operatorname{dist}(\cdot, C)$ is continuous, and the sign changes only across points of C, so f is continuous.

Hint 3. Every neighbourhood of $x \in C$ contains a whole step-k piece of the construction containing x, and that piece contains removed intervals of levels $k+1$ and $k+2$, i.e. of both signs. So f crosses the axis at every point of the uncountable set C.

Problem 7

(a)

Hint 1. Let $g(x_0) = \lim_{x \to x_0} f(x)$. If $f(x_0) > g(x_0)$, then $f(x) < f(x_0)$ for all $x \neq x_0$ near x_0, so x_0 is a local maximum. The case $f(x_0) < g(x_0)$ is similar.

(b)

Hint 1. By a), f is continuous, so by Theorem 2 it is enough to show that f is injective.

Hint 2. If $f(a) = f(b)$ with $a < b$, then either f is constant on $[a, b]$, or its maximum or minimum on $[a, b]$ differs from $f(a)$ and is attained inside. In both cases there is a local extremum.

Problem 8

(a)

Hint 1. $f([0, a])$ is a compact segment $[m, M] \subset (0, 1)$. Values in $(0, m)$ and $(M, 1)$ must therefore be taken on $(a, 1)$.

Hint 2. So $f((a, 1))$ is an interval containing values arbitrarily close to 0 and to 1; hence it is $(0, 1)$.

(b)

Hint 1. You need oscillations whose amplitude tends to $\frac{1}{2}$ without reaching it, e.g. $f(x) = \frac{1}{2} + \frac{x}{2} \sin \frac{1}{1-x}$.

Problem 9

Hint 1. Key observation: if $0 < s < t$ and $t/s = p/m$ is rational, then applying the hypothesis with $x = s/m$ gives $f(s) = f(m \cdot \frac{s}{m}) \leqslant f(p \cdot \frac{s}{m}) = f(t)$. Also $f(0) \leqslant f(x)$.

(a)

Hint 1. Answer: no. Rational and irrational points never appear in the same sequence (nx), and continuity on $[0, 1]$ says nothing beyond 1.

Hint 2. Take $f(x) = x$ for $x \in [0,1]$ and for rational x, and $f(x) = x + 1$ for irrational $x > 1$. Then every sequence $f(nx)$ is increasing, but $f(\sqrt{2}) > f(\frac{3}{2})$.

(b)

Hint 1. Answer: yes. For $0 < s < t$ approximate t by points $t_k \to t$ with t_k/s rational, and pass to the limit in $f(s) \leqslant f(t_k)$ using continuity.

Problem 10

Hint 1. Let $f(p) = a$ and $f(q) = b$, say $p < q$ (the other case is symmetric). Shrink $[p, q]$ to a segment on which the values a and b are taken only at the endpoints.

Hint 2. Let p' be the last point in $[p, q]$ with $f = a$, and q' the first point in $[p', q]$ with $f = b$. On (p', q'), f takes neither a nor b, so by the IVT it stays in (a, b). Hence $f([p', q']) = [a, b]$.